Christian Camilo Gaviria Castro



Sr. Cloud & Security Reliability Engineer & Technical Leader · DevSecOps · Kubernetes · AWS · Cloud-Native · Linux · Windows · Platform Engineering · GitOps · Terraform · IPv6

me@cgaviria.com · +57 300 343 7758 · linkedin.com/in/cgaviria44 · linktr.ee/cgaviria44 · Medellín, Colombia (GMT-5)

PROFILE

Sr. Cloud & Security Reliability Engineer and technical leader with 10+ years of experience designing, automating, and scaling cloud-native infrastructure for production environments. Specialized in Kubernetes, AWS, and GNU/Linux, with a strong background in networking, cloud security, and DevOps/DevSecOps practices.

Currently a **Cloud Specialist / Cloud Engineer** on the DevOps team at **GFT Technologies**, building and automating cloud infrastructure for financial-services platforms. Previously at **Cuemby** (through May 2026), led the reliability engineering and platform strategy for a hybrid cloud serving businesses across the Americas — driving high availability, infrastructure automation, and platform observability at scale.

For 4 years, I led a research group on networking and communications security at ITM (Instituto Tecnológico Metropolitano), building a deep technical foundation and a passion for knowledge sharing and mentorship that I bring to every team I work with.

KEY ACHIEVEMENTS

- Architect and operate production hybrid cloud at US-based Cuemby — public (AWS, GCP, Azure, CIVO, Equinix Metal), private cloud and on-premise baremetal; incident response with bilingual EN/ES client liaison.
- Production Kubernetes at telco scale — Cilium (eBPF networking), Rook Ceph (distributed storage), Velero (backup, DR & DRP), KubeVirt; cluster lifecycle and security hardening.
- Deployed 5G and cellular core network functions on Kubernetes at production scale — running containerized network functions (CNFs) for high-performance workloads.
- Cloud-native edge optimization with Unikraft unikernels — deploying serverless applications as lightweight, single-purpose VMs for extreme performance at the edge.
- A decade architecting ISP and MSP telecom networks — BGP, IPv6 (LACNIC), full OSI stack, GPON deployments across Huawei, ZTE, TP-Link and CData, plus IPTV and VoIP for end users.

— Led cloud migrations for clients across the Americas — monolithic to elastic AWS architectures using Terraform/Ansible IaC and blue/green deployments for zero-downtime cutovers.

— Promoter of open-source technologies to cut operational costs — Kubernetes + Cilium + Ceph + Velero replacing proprietary alternatives; KVM, Proxmox, Apache CloudStack as cost-effective virtualization.

EXPERIENCE

Cloud Specialist / Cloud Engineer — DevOps

Jul 2026 — Present

GFT Technologies · Hybrid · Medellín, Colombia

— Cloud Specialist / Cloud Engineer within the DevOps team, currently on the "Garantías y Cartera" (guarantees & loan-portfolio) initiative for financial-services workloads.

— Design, automate and operate cloud infrastructure and CI/CD pipelines to support reliable, secure delivery of banking applications.

— Apply Infrastructure as Code (Terraform), GitOps and containerization/Kubernetes practices to standardize environments and reduce deployment risk.

— Collaborate with development and platform teams on observability, security and reliability across the delivery lifecycle.

Senior Site Reliability Engineer — Cloud & Security

Apr 2023 — May 2026 · 3y 1m

Cuemby · Remote · US-based

— Lead the architecture and operation of hybrid cloud platforms — public (AWS, GCP, Azure, CIVO, Equinix Metal), private cloud and on-premise baremetal — driving decisions on provisioning, network design, observability, security and incident response.

— Lead advanced Kubernetes workloads in production across multiple distributions including Talos Linux: Cilium-based networking (eBPF), Rook Ceph distributed storage, Velero backups, disaster recovery (DR) and disaster recovery planning (DRP), ingress, service mesh and load balancers, cluster lifecycle, etcd backups and security hardening.

— Deployed 5G and cellular core network functions on Kubernetes at production scale, running containerized network functions (CNFs) for high-performance workloads.

— Cloud-native edge optimization with Unikraft unikernels — packaging and deploying serverless applications as lightweight, single-purpose VMs for extreme performance at the edge.

— Lead virtualization strategy across modern (KubeVirt, OpenStack) and traditional stacks (KVM, Proxmox, VMware, Apache CloudStack); designed and automated baremetal installation and mass OS provisioning pipelines using MAAS (Canonical), Tinkerbell and Seeder Harvester across data centers.

— Built reusable IaC modules with Terraform, CloudFormation and Ansible, and constructed CI/CD pipelines (Jenkins, GitHub Actions, AWS CodePipeline/CodeBuild) — including rollback and blue/green deployments; applied Git best practices with GitHub and GitLab for versioning, code review and change control

across all IaC repositories; maintained technical documentation using GitHub Issues and Projects, and produced SWOT analysis matrices for infrastructure decisions.

- Automated operational and infrastructure workflows with Ansible across multi-environment deployments; delivered a Kubernetes automation project for the Colombian Air Force (FAC), covering deployment pipelines and cluster lifecycle management.
- Developed Python and Bash scripts to automate routine Linux tasks — reducing manual toil and improving operational consistency across environments.
- Design data resilience strategy — distributed and software-defined storage (Rook Ceph, TrueNAS, OpenEBS, Mayastor), object and block storage with MinIO, clustering, RAID, scheduled backups and disaster-recovery plans across multi-region setups.
- Lead observability strategy across distributed workloads — metrics (Prometheus, Grafana), logs (Loki), traces (OpenTelemetry, Jaeger, Tempo), alerting (AlertManager → Slack, Teams, PagerDuty) — and drive root-cause analysis for production incidents.
- Drive DevSecOps practices: internal security audits (Wazuh), SSL/TLS, IAM hardening, secrets scanning (GitGuardian), container vulnerability scanning (Trivy), Kubernetes security benchmarking (Kube-bench), Cloudflare WAF and CDN, exposed port detection, and container image security across repositories and production environments.
- Integrate AI tooling (Claude Code, ChatGPT) with structured prompt engineering into runbooks, IaC reviews and postmortems; lead architecture negotiations and live incident response with US-based clients as bilingual (EN/ES) technical liaison.
- Implemented Vanta for compliance automation — centralizing security controls, evidence collection and continuous compliance monitoring across infrastructure and engineering processes.
- Managed network fabric and BGP automation with Netris across bare-metal deployments on Equinix Metal; configured and operated load balancers for Kubernetes and on-premise environments.
- Deployed Telepresence for seamless remote developer access to production Kubernetes clusters; installed, configured and administered Nextcloud as a self-hosted collaboration and file-sharing platform.
- Contributed to the development of an internal marketplace integrating top community tools with one-click install — including n8n, OpenClaw, MySQL, PostgreSQL, WordPress, and others — enabling clients to deploy production-ready applications instantly on the platform.
- Produced video tutorials for installing and setting up marketplace applications — including AI tools (n8n, OpenClaw), WordPress, MySQL, and others — guiding clients through deploying and configuring applications on the platform.

Cloud Engineer

Oct 2019 — Dec 2025 · 6y 2m · project-based

Negocios NEX · Remote

- Designed and implemented cloud network architectures on AWS for digital products.
- Led migration and optimization of legacy infrastructure to AWS, focusing on cost, resilience and security.
- Database administration across PostgreSQL, MySQL, SQL Server and Amazon Aurora.

— Conducted infrastructure and security audits for client companies, identifying vulnerabilities and delivering actionable remediation plans.

— Applied version control best practices with GitHub and GitLab for Ansible and Terraform codebases — enforcing branching strategies, code review workflows and change traceability.

— Designed and implemented backup and disaster recovery (DR) strategies for client cloud workloads — including scheduled backups, point-in-time restoration and multi-region resilience planning.

Infrastructure Engineer

2019 — Dec 2025 · ~6y · freelance

Smart Town SAS · Remote

— Design, deploy, and manage the full ISP technology stack — dual-stack IPv4/IPv6 networking, security hardening, and day-to-day operations of the service provider infrastructure.

— Administer GPON fiber-optic infrastructure for FTTH delivery — OLT/ONT provisioning, troubleshooting and capacity planning.

— Design and operate wireless radio link networks for distribution and last-mile connectivity.

— Provide technical consulting to client companies on network architecture, ISP infrastructure modernization and security posture.

— Deployed and administered a DNS server with DNSSEC validation and privacy-preserving resolution protocols — DNS over HTTPS (DoH) and DNS over TLS (DoT) — for the ISP infrastructure.

Infrastructure Engineer

Mar 2021 — Dec 2023 · 2y 9m · project-based

TikAcademy · Remote

— Consulted for ISPs and enterprises on modernization of their network stack using MikroTik RouterOS.

— Translated business requirements into technical roadmaps for non-technical stakeholders.

Cloud Engineer

Jan 2022 — Jul 2023 · 1y 7m

Droptek · Remote · Argentina-based

— Planned and executed the migration of monolithic systems to elastic, autoscaling AWS architectures.

— Authored Terraform and CloudFormation templates to make environments reproducible and reduce drift.

— Designed and implemented production-grade elastic deployments in AWS for customer workloads.

Sysadmin · Network · Wireless · Cloud Engineer

Sep 2019 — Dec 2021 · 2y 4m

EJ Soluciones · Medellín, Colombia

— Designed wired networks, enterprise Wi-Fi and CCTV / IP-surveillance systems for hotels and corporate customers.

— Deployed and operated MSP-scale environments with MikroTik, UniFi, Cisco and Elastix/Issabel PBX (VoIP).

— Trained internal technicians on network device configuration and best practices.

— Managed VMware vCenter environments — handling virtual machine migrations, infrastructure upgrades and backup configuration for virtualized workloads.

— Administered Windows Server environments and Active Directory — DNS, Group Policy Objects (GPOs), access controls, disk quotas, remote access (RDP/VPN) and Active Directory migrations for corporate clients.

— Implemented Microsoft security hardening — firewall policies, endpoint controls, Microsoft Security baselines and Entra ID (Azure AD) identity management across corporate client environments.

IT Support Coordinator

Aug 2016 — Dec 2021 · 5y 5m

PEG Soluciones – Teltic · On-site & Remote · Antioquia, Colombia

— Led the physical and logical design of telecommunications networks for ISP clients, including GPON deployments with Huawei, ZTE, TP-Link and CData OLTs/ONTs.

— Led the configuration and operation of core network devices (BGP routers, switches, OLTs) plus IP telephony (VoIP) and IPTV services.

— Led technical documentation efforts and mentored support technicians on MikroTik, Ubiquiti and major-vendor GPON equipment.

— Conducted network and infrastructure audits for client companies, assessing equipment, topology and security posture to identify gaps and recommend improvements.

— Designed and operated CGNAT (Carrier-Grade NAT) solutions for ISP address management — configuring port block allocations to expose client network services while conserving public IPv4 resources.

Network Administrator & Researcher — Information Security Group (SISSI)

Aug 2016 — Oct 2021 · 5y 3m

ITM — Instituto Tecnológico Metropolitano · On-site & Remote · Medellín, Colombia

— Led and coordinated the Network Security and IoT Security research track (OSI model), defining the research agenda and guiding the group's technical direction; also served as assistant on security incident management and digital forensic analysis (Feb 2018 — Oct 2021).

— Served as technical lead for network administration within the Information Security research group, supporting and enabling research activities (Aug 2016 — Oct 2021).

TECHNICAL PROFICIENCY

Networking	BGP · IPv6 · GPON · ISP/MSP	10/10
5G	K8s-native · core network	5/10
Unikraft	Edge · serverless VMs	6/10
Kubernetes	KCNA · Cilium · Ceph · Velero	9/10

AWS	Solutions Architect Associate	9/10
IaC	Terraform · Ansible · CloudFormation	9/10
Linux & Virt.	KVM · Proxmox · VMware	9/10
Multi-cloud	AWS · GCP · Azure · CIVO · Equinix	8/10
CI/CD	Jenkins · GitHub Actions · AWS	8/10
Observability	Prometheus · Grafana · Loki · OTel	8/10
DevSecOps	GitGuardian · IAM · SSL/TLS	8/10
Scripting	Python · Bash (Linux automation)	5/10

FULL STACK

cloud	AWS (Solutions Architect Associate, Cloud Practitioner), GCP, Azure, CIVO, Equinix Metal, DigitalOcean; hybrid, private and public cloud
kubernetes	KCNA (CNCF), Talos Linux, Cilium (eBPF CNI), Rook Ceph, Velero, KubeVirt, Helm, ingress controllers, service mesh, load balancers, Telepresence, cluster lifecycle, etcd backups
containers	Docker, Docker Compose, containerd, multi-stage builds
virtualization	KVM, Proxmox, VMware vCenter, KubeVirt, OpenStack, Apache CloudStack
iac	Terraform, Ansible, CloudFormation, cloud-init, PXE provisioning, MAAS (Canonical), Tinkerbell, Seeder Harvester
ci/cd	Jenkins, GitHub Actions, GitLab CI, AWS CodePipeline/CodeBuild/CodeDeploy, blue/green, canary, zero-downtime migrations
source control	GitHub, GitLab, Bitbucket, Git workflows
atlassian	Jira, Confluence, ticket hygiene, technical documentation
observability	Prometheus, Grafana, Loki, Jaeger, Tempo, OpenTelemetry, AlertManager (Slack · Teams · PagerDuty)
security	Firewalls, UTM, WAF (Cloudflare), VPNs (site-to-site, client-to-site, IPsec, WireGuard, OpenVPN, Tailscale, mesh), Wazuh (SIEM/audits), Trivy, Kube-bench, GitGuardian, SSL/TLS, IAM hardening, Cisco Cybersecurity Essentials
compliance	Vanta (SOC 2, security posture, continuous compliance, evidence automation)

storage & dr/drp	Rook Ceph, TrueNAS, OpenEBS, Mayastor, MinIO (object & block storage), RAID, clustering, scheduled backups, Velero, disaster recovery (DR), disaster recovery planning (DRP), multi-region
self-hosted	Nextcloud (installation, configuration, administration)
networking	BGP, OSPF, IPv4/IPv6, DNS/DNSSEC/DoH/DoT, full OSI stack, Wireshark, tcpdump, Netris (network fabric), MikroTik (MTCNA · MTCTCE · MTCSE · MTCIPv6E), Cisco CCNA, Ubiquiti UniFi
wireless	Enterprise Wi-Fi, MikroTik Hotspot, CCTV / IP surveillance
telco	5G core on Kubernetes, Unikraft unikernels, GPON (Huawei, ZTE, TP-Link, CData), IPTV, IP telephony (Asterisk, Elastix, Issabel)
databases	PostgreSQL, MySQL, SQL Server, Amazon Aurora (administration, backups, performance)
os	Linux server administration, Windows server
scripting	Python, Bash — task automation and basic Linux scripting
ai engineering	Claude Code, ChatGPT, GitHub Copilot, structured prompt engineering, AI-assisted runbooks, IaC review, postmortems

EDUCATION

Engineer's degree, Telecommunications
ITM · 2016–2019

Technologist's degree, Telecommunications
ITM · 2013–2016

LANGUAGES

Spanish — Native

English — B2 — Upper Intermediate
A1 & A2 certified · B1 in progress · EF SET 59/100 (B2) · 3+ years at Cuemby (EE.UU.)

CERTIFICATIONS

cloud	AWS Certified Solutions Architect — Associate (SAA-C03), AWS Certified Cloud Practitioner (CLF), Open-Source Technologies on Microsoft Azure
kubernetes	Kubernetes & Cloud Native Associate (KCNA, CNCF)
observability	LFS148 — Getting Started with OpenTelemetry
networking	MikroTik MTCNA · MTCTCE · MTCSE · MTCIPv6E · LACNIC IPv6 Advanced · Cisco CCNA Industrial
networking	LACNIC — Interconnection of Autonomous Systems, Public & Private — 2026 Edition (BGP & IPv6)

security	Cisco Cybersecurity Essentials
devops / iac	Infrastructure as Code with Terraform, Docker, AWS Storage (Platzi)
methodology	Scrum Foundation Professional (SFPC)
networking	IPv6 Certification — Hurricane Electric (2022)
open source	Festival Latinoamericano de Instalación de Software Libre — Speaker (FLISOL, 2022)
productivity	Microsoft Office Tools: Microsoft Excel — SENA (2010)

HONORS & AWARDS

- **2020** — ITM Engineering Week, Graduate recognition footprint (Egresado Huella)
- **2022** — Speaker at FLISOL Medellín
- **2025** — Speaker at KCD Colombia 2025
- **LACNIC 37** — Participation

TALKS

From Bare Metal to Cloud: Automate Your Infrastructure with MAAS and Harvester

KCD Colombia 2025

Live demo on deploying and orchestrating physical servers as cloud-like resources using MAAS and Harvester. Covering edge computing, hybrid environments, and 100% open source infrastructure automation.

Docker & Cloud

FLISOL Medellín · April 2022

Talk on containerization with Docker and cloud fundamentals for the open source community at the Latin American Free Software Installation Festival.

SOFT SKILLS

critical thinking	Process optimization, reasoning under pressure
leadership	Decision-making, influence, technical mentorship
business perspective	Client-facing communication, requirement translation
negotiation	Stakeholder communication across technical and non-technical audiences

product thinking	Continuous improvement mindset, ideas-to-impact
bilingual	EN / ES collaboration across distributed teams

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